1. Literature Review

Smart Liquid Tank Monitoring Systems

1.1 Overview

Smart liquid tank monitoring systems has become important in water management and consumption in agriculture and manufacturing also chemical processing and industrial storage. These systems uses sensors and electronic control units to measure and know the physical condition of any stored liquids by integrated measurements such as liquid level and mass and temperature a monitoring system will provide more complex info about tank condition than a system that measures only one parameter. integration of these measurements make the system able to estimate liquid volume and fill percentage and average density and changes in liquid quantity over time. microcontroller or industrial control system can process the collected data and display real time measurement through a graph interface. The system can also make warnings when measurements reach limits. Industrial tank gauging solutions shows the importance of combining level and temperature to support management and process monitoring.

The proposed smart liquid tank monitoring system follows this approach by combining multiple sensors to monitors physical properties of liquid. the system aim to improve the accuracy also make continuous observation, identify unusual measurement patterns and provide useful information for operational decision-making. By comparing different measured parameters, the system can also help identify potential sensor errors and unexpected changes in tank conditions.

1.2 Historical Background

Traditional liquid storage tank commonly relied on manual level indicator or visual inspection and individual measuring instruments to determine the quantity of gauged liquid. these methods can provide useful info but they may require regular operator intervention and may not provide continuous information about all tank conditions.The development of industrial instrumentation introduced automatic level gauges and temperature sensors and weighing systems and electronic monitoring equipment. These technologies enabled more consistent measurement and reduced the need for manual inspection. The International Organization of Legal Metrology (OIML) established technical recommendations for automatic level gauges used to measure liquid levels in stationary storage tanks. These recommendations address measurement performance and the use of level readings with tank calibration tables to determine stored liquid volume (OIML, 2008).

Modern industrial tank-gauging systems extend these measurement capabilities by combining several instruments and connecting them to inventory management systems. For example, Endress+Hauser provides tank-gauging solutions that integrate level measurement, temperature monitoring, and inventory management for applications in industries such as oil and gas, chemicals, and food production (Endress+Hauser, n.d.).

These developments provide a foundation for smart liquid tank monitoring systems that combine physical measurements, electronic data processing, and real-time visualization. A university-level prototype can apply

the same fundamental principles using lower-cost sensors and a microcontroller while providing opportunities to investigate measurement accuracy, calibration, and data integration.

1.3 Recent Trends

Recent developments in liquid tank monitoring emphasize the integration of multiple sensors, real-time data acquisition, remote monitoring, and automated data analysis. Level sensors based on ultrasonic, radar, hydrostatic, and capacitive measurement principles can be selected according to the tank geometry, liquid properties, operating environment, and required measurement performance. The selection of a suitable sensor is important because temperature, material properties, installation conditions, and tank geometry can influence measurement results (Endress+Hauser, n.d.).

Another important trend is the integration of level measurements with mass and temperature measurements. When the tank geometry is known, the measured liquid level can be converted into volume. If the net mass of the liquid is also measured, its average density can be estimated. Temperature measurements provide additional information because the density and volume of many liquids change with temperature. These relationships are important when comparing measurements obtained under different operating conditions.

Industrial Internet of Things (IIoT) technologies have also expanded the possibilities for collecting, transmitting, and analyzing sensor data. Connected monitoring systems can send information to control systems, remote dashboards, or industrial software platforms. A literature review by Liu et al. (2023) discusses remote monitoring and maintenance in industrial environments and describes the role of industrial internet technologies in detecting abnormalities and supporting operational decisions.

For smart tank monitoring, these developments make it possible to record changes in liquid level, mass, and temperature continuously. The resulting data can be used to investigate filling and draining rates, identify inconsistent sensor readings, and generate warnings when operating conditions fall outside predefined limits.

1.4 Main Future Directions

Future developments in smart liquid tank monitoring are expected to focus on improving measurement accuracy, communication, automation, and data analysis. One important direction is the integration of wireless communication technologies that allow operators to monitor tank conditions remotely. Wireless connectivity can reduce the need for direct access to individual tanks, particularly when tanks are distributed across a large facility.

Another direction is the improvement of sensor calibration and measurement uncertainty estimation. Incorrect sensor installation, inaccurate calibration, tank deformation, and unsuitable tank-volume calculations can introduce errors into estimated liquid quantities. Research published by OIML authors highlights the importance of correctly linking automatic level-gauge readings with tank calibration tables, since errors in this process can produce systematic biases in calculated liquid volume and mass (Samoilenko et al., 2024).

Advanced monitoring systems may also incorporate data filtering, anomaly detection, predictive maintenance, and automatic pump control. By analyzing historical measurements, a monitoring system could identify

unusual filling patterns, detect possible leaks, or recognize discrepancies between measured mass and estimated volume. However, these functions require appropriate validation before being relied upon for safety-critical decisions.

Future prototypes can also incorporate graphical visualization, data logging, configurable alarms, and communication with external control systems. These improvements would make a low-cost monitoring prototype more representative of industrial systems while preserving its value for educational experiments and engineering research.

1.5 Main Applications of Smart Liquid Tank Monitoring

Smart liquid tank monitoring systems have applications in several sectors. In water storage and distribution, continuous level monitoring helps operators maintain adequate supplies and reduce the risk of overflow or empty tanks. In agriculture, monitoring water storage can support irrigation planning and improve the management of available resources.

In manufacturing and chemical processing, the measurement of liquid quantity and temperature supports inventory control, material consumption and the maintenance of suitable operating conditions. In food and beverage production, monitoring can help operators observe storage conditions and manage liquid ingredients throughout production processes.

Laboratory applications provide an additional opportunity to investigate the relationship between mass, volume, density, and temperature. A prototype equipped with a level sensor, weighing system, and temperature sensor can be used to compare calculated physical properties with reference measurements. Such experiments can help evaluate sensor performance and determine the effects of calibration errors and measurement uncertainty.

The suitability of a monitoring system depends on the liquid being measured, the tank design, the required accuracy, and the environmental conditions. Industrial installations may require

specialized sensors, compatible construction materials, and appropriate protection against chemical exposure or other operational hazards.